

Project: Event-Driven Cloud Infrastructure Cost Optimization & FinOps Automation

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Overview

I designed and deployed a fully automated, serverless cloud cost optimization system using Terraform to eliminate silent infrastructure waste.

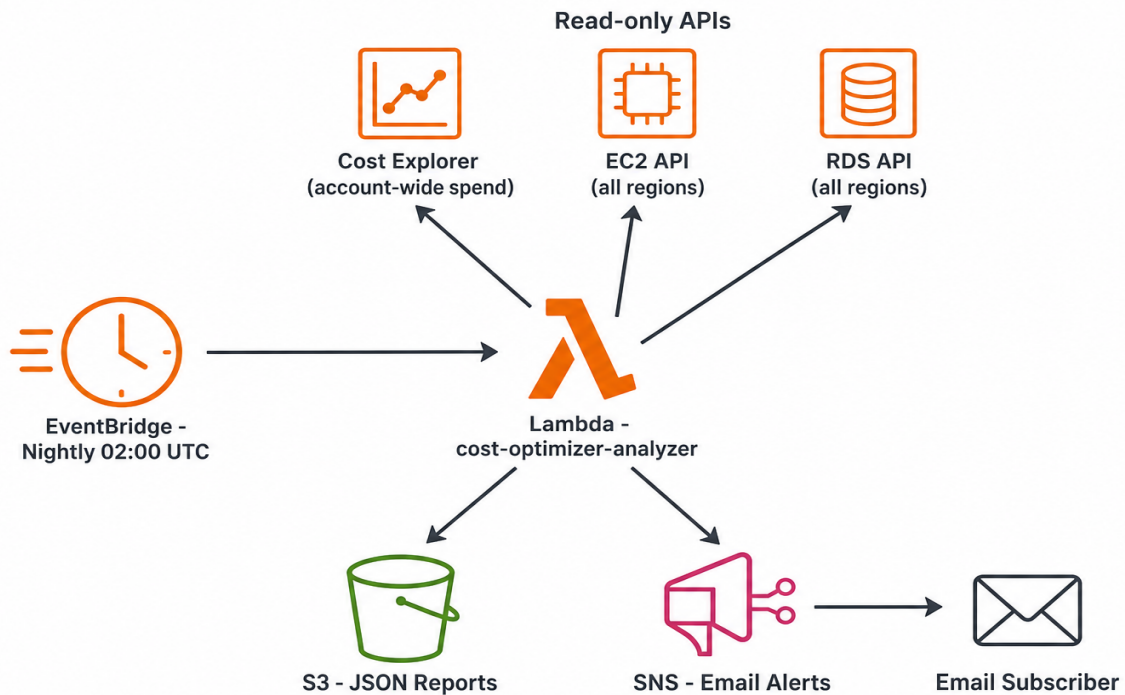
Imagine running an office building where employees leave the lights on, the AC blasting, and rental cars ticking in metered spots every single night—that is exactly what happens in the cloud when teams forget to delete temporary setups, leaking thousands of dollars. To solve this, I built an event-driven system that wakes up every night at 2:00 AM UTC, scans the entire global cloud infrastructure for unlinked or abandoned resources, and emails a clean, actionable report before morning coffee.

Operational Flow

Here is exactly what happens automatically at 2:00 AM UTC while the team is completely offline:

- 1. The Trigger:** An Amazon EventBridge schedule wakes up the Python-based AWS Lambda function.
- 2. The Scan:** The function queries the AWS Cost Explorer API and scans all enabled AWS regions for hidden waste.
- 3. The Audit:** It actively hunts down unattached EBS volumes, unused Elastic IPs, stopped EC2/RDS instances, and orphaned snapshots.
- 4. The Delivery:** The system drops a detailed raw JSON report into an Amazon S3 bucket and simultaneously fires a clean, actionable executive summary straight to the user's inbox via Amazon SNS.

Architecture



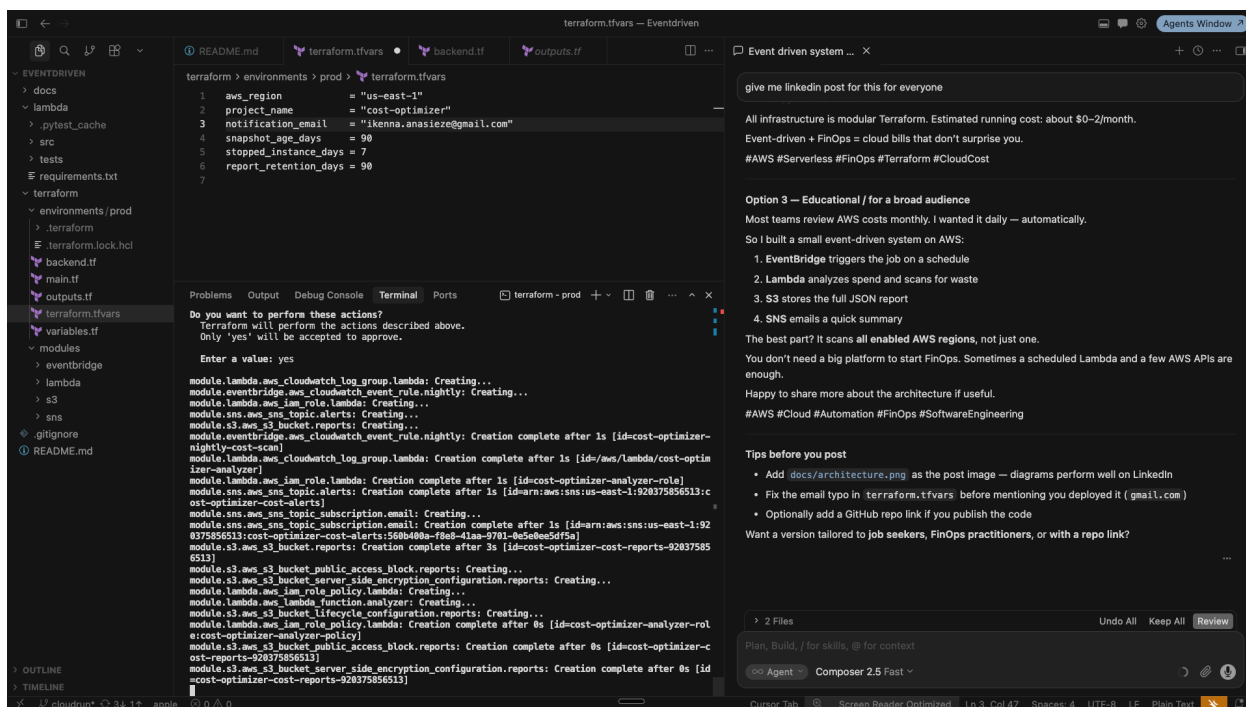
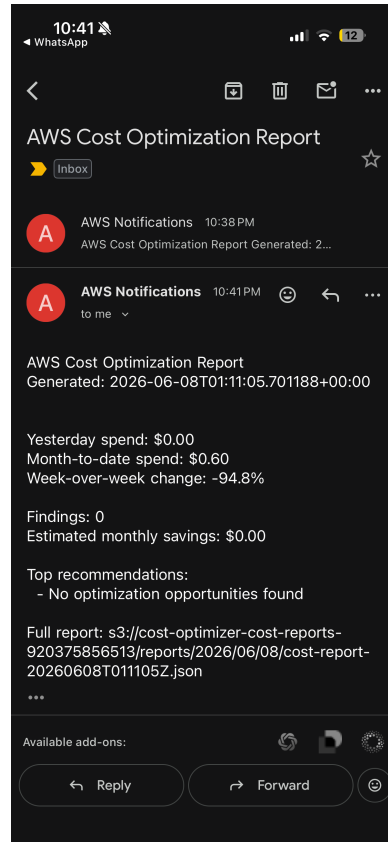
- **Event Bridge** -> Serverless cron scheduler to trigger nightly audits.
- **AWS Lambda** -> Python 3.12 runtime executing resource scans and cost analysis.
- **AWS APIs** -> Cost Explorer, EC2, RDS – Programmatic deep-dive into active infrastructure.
- **Amazon S3** -> Secure, centralized storage for detailed JSON audit reports.
- **Amazon SNS** -> Direct alerting pipeline for executive summaries.
- **Terraform** -> 100% modular Infrastructure as Code (IaC) for rapid, reproducible deployments.

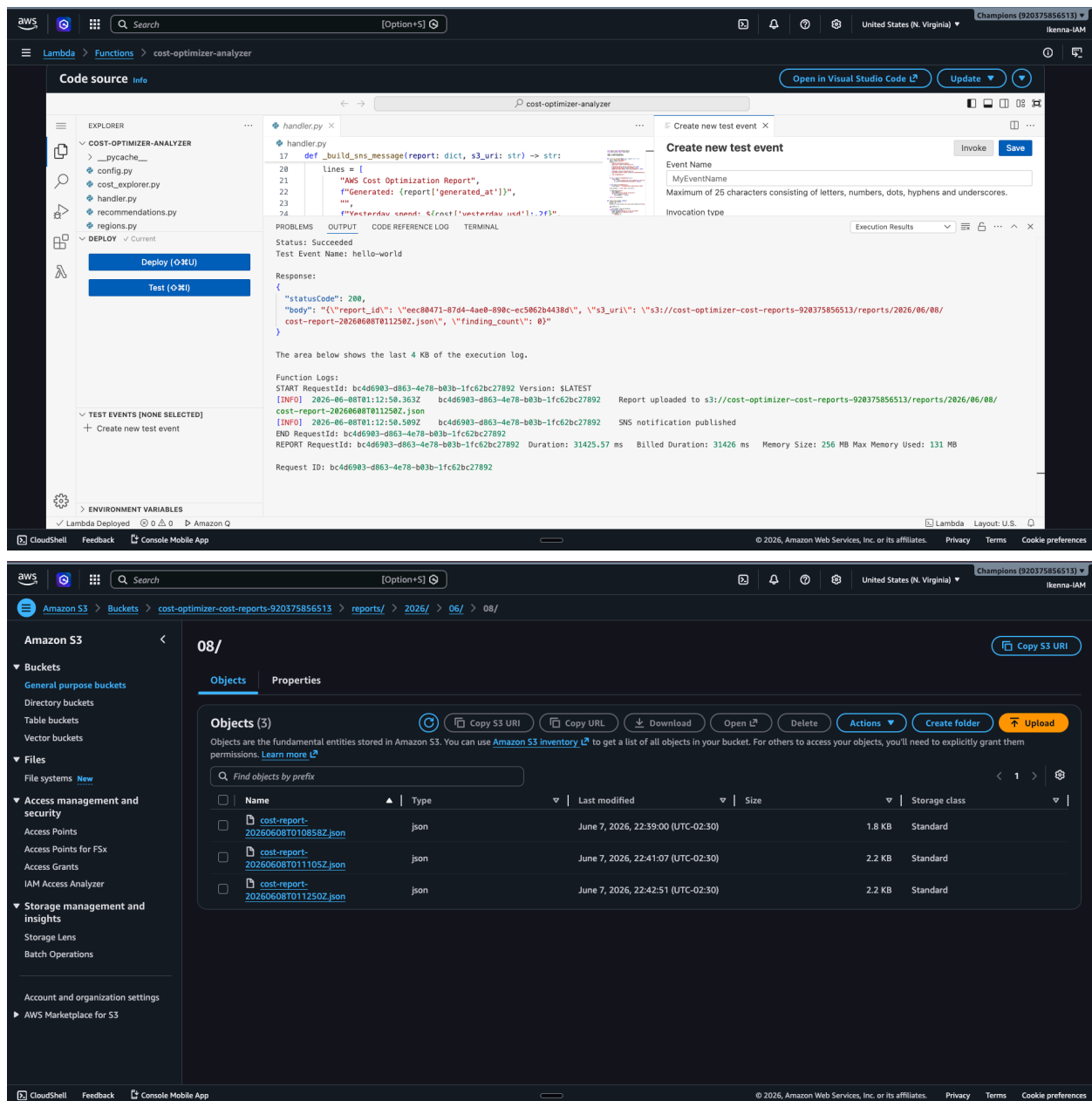
Key Innovation

This system leverages the pure power of Event-Driven Architecture combined with structured FinOps practices, allowing organizations to:

- **Eliminate Manual Overhead:** Replaces error-prone manual environment checks and messy spreadsheets with a hands-off, automated routine.
- **Achieve Multi-Region Visibility:** Programmatically audits all enabled global cloud regions simultaneously, ensuring zero blind spots.

- **Optimize Operational Costs:** Runs on a 100% serverless execution layer, meaning the automation itself costs next to nothing to run.
- **Deliver High-ROI Visibility:** Connects technical resource states directly to cost metrics, providing immediate financial insights with zero ongoing operational effort.





Impact

- **100% Automated Governance:** Shifted from manual infrastructure tracking to a zero-touch daily audit.
- **Instant Cost Leakage Visibility:** Immediate identification of low-hanging waste (orphaned volumes, idle instances, and unattached IPs).
- **Zero Operational Maintenance:** Native serverless design requires zero server management or patching.
- **Rapid Deployment Time:** Built entirely with modular Terraform, allowing the entire stack to be spun up or integrated into new AWS accounts in minutes.